

NCERT Exemplar Solutions for Class 12

Chemistry

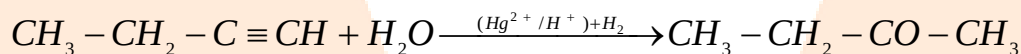
Chapter 12 – Aldehydes, Ketones and Carboxylic Acids

I. Multiple Choice Questions (Type-I)

1. Addition of water to alkynes occurs in acidic medium and in the presence of Hg^{2+} ions as a catalyst. Which of the following products will be formed in addition to water to but-1-yne under these conditions?

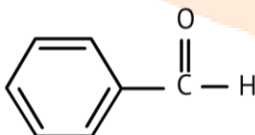
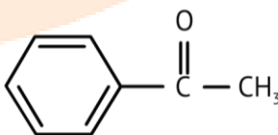
- (i) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \overset{\text{O}}{\parallel}{\text{C}} - \text{H}$ (ii) $\text{CH}_3 - \text{CH}_2 - \overset{\text{O}}{\parallel}{\text{C}} - \text{CH}_3$
- (iii) $\text{CH}_3 - \text{CH}_2 - \overset{\text{O}}{\parallel}{\text{C}} - \text{OH} + \text{CO}_2$ (iv) $\text{CH}_3 - \overset{\text{O}}{\parallel}{\text{C}} - \text{OH} + \text{H} - \overset{\text{O}}{\parallel}{\text{C}} - \text{H}$

Ans: Correct option: B



Butan-2-one is produced when butyne is hydrated in the presence of Hg ions as a catalyst. The ketone is the product. After the triple bond is broken, the OH group attaches to the secondary carbocation.

2. Which of the following compounds is most reactive towards nucleophilic addition reactions?

- (i) $\text{CH}_3 - \overset{\text{O}}{\parallel}{\text{C}} - \text{H}$ (ii) $\text{CH}_3 - \overset{\text{O}}{\parallel}{\text{C}} - \text{CH}_3$
- (iii)  (iv) 

Ans: Correct option: (ii)

In nucleophilic addition processes, the reactivity of aldehydes and ketones are steric factors. The carbonyl group with the highest positive charge will be the most reactive, and if these groups have a positive inductive impact, the attack tendency will be reduced.

3. The correct order of increasing acidic strength is _____.

A. Phenol < Ethanol < Chloroacetic acid < Acetic acid

B. Ethanol < Phenol < Chloroacetic acid < Acetic acid

C. Ethanol < Phenol < Acetic acid < Chloroacetic acid

D. Chloroacetic acid < Acetic acid < Phenol < Ethanol

Ans: Correct option: C

Because the phenoxide ion produced is stabilised by resonance, phenol is more acidic than ethanol. When it comes to ethanol, resonance does not help to stabilise the phenoxide ion. The acidity of chloroacetic acid is higher than that of acetic acid. Alcohols and phenols are less acidic than carboxylic acids.

4. Compound $\text{Ph}-\text{O}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Ph}$ can be prepared by the reaction of _____.

A. Phenol and benzoic acid in the presence of NaOH

B. Phenol and benzoyl chloride in the presence of pyridine

C. Phenol and benzoyl chloride in the presence of ZnCl_2

D. Phenol and benzaldehyde in the presence of palladium

Ans: Correct option: B

When phenol reacts with acid chloride in the presence of an alkali, the alkali is pyridine, and the acid chloride is benzoyl chloride, compound $\text{C}_6\text{H}_5\text{OCOC}_6\text{H}_5$ is formed. During this process, pyridine is utilised to neutralise 4HCl .

5. The reagent which does not react with both acetone and benzaldehyde.

A. Sodium hydrogen sulphite

B. Phenyl hydrazine

C. Fehling's solution

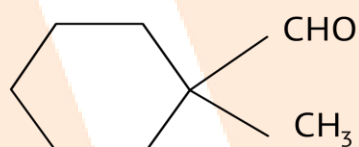
D. Grignard reagent

Ans: Correct option: C

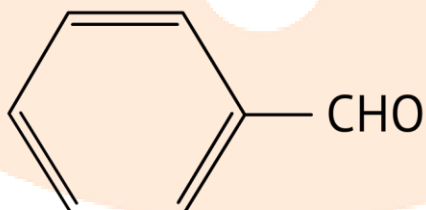
Acetones belong to the ketonic functional category. To distinguish between aldehyde and ketone, utilise Fehling's solution. Carbonyl groups are found in sodium hydrogen sulphite, phenyl hydrazine, and Grignard reagent. Ketones and aromatic aldehydes do not react with Fehling's solution.

6. Cannizaro's reaction is not given by _____.

A



B



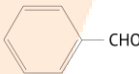
C. HCHO

D. CH_3CHO

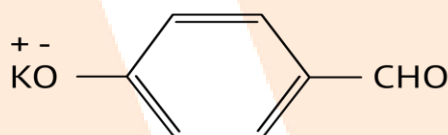
Ans: Correct options: D

Aldehydes without α - hydrogen atom are the subject of Cannizaro's reaction.

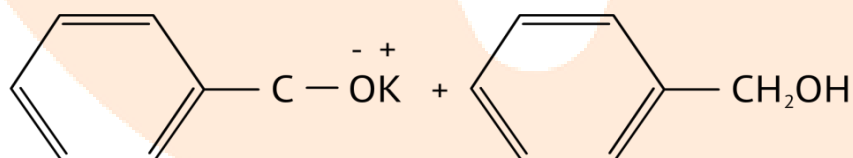
Option D, CH_3CHO has 3 α hydrogen atoms and it does not give Cannizaro's reaction

7. Which product is formed when the compound  is treated with concentrated aqueous KOH solution?

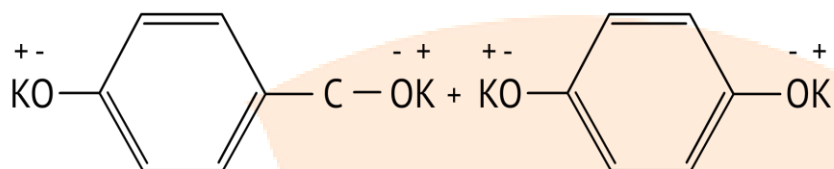
A.



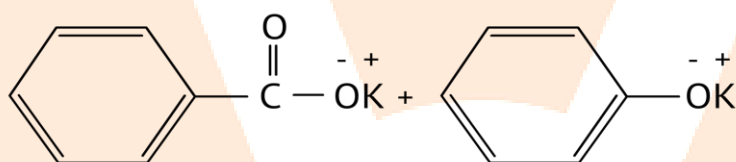
B



C.

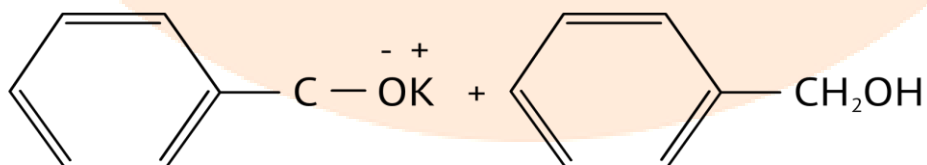


D.

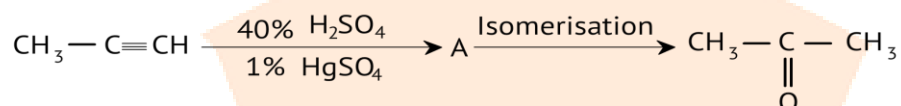


Ans: Correct option: B

In KOH adds to the compound



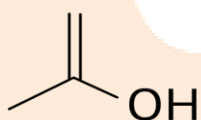
Due to the production of a stable conjugate base after hydronium is removed from phenol, it is more stable than alcohol.



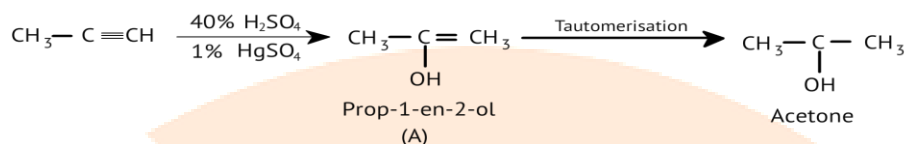
8. Structure of 'A' and type of isomerism in the above reaction are respectively

- A. Prop-1-en-2-ol, metamerism
- B. Prop-1-en-1-ol, tautomerism
- C. Prop-2-en-2-ol, geometrical isomerism
- D. Prop-1-en-2-ol, tautomerism

Ans: Correct option: D

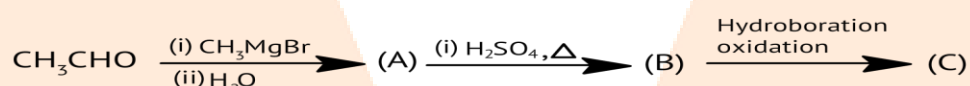


Prop-1-en-2-ol



Tautomers are structural isomers that exist in two or more interconvertible structures of a single chemical molecule. Tautomerism transforms 'A' into acetone.

9. Compounds A and C in the following reaction are _____.

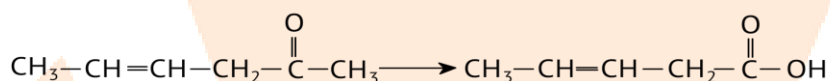


- A. identical**
- B. positional isomers**
- C. functional isomers**
- D. optical isomers**

Ans: Correct option: B

The position of a functional group in the same carbon chain differs in positional isomerism. The functional group in the chemical or the unsaturated double or triple bond in the compound can help us figure it out.

10. Which is the most suitable reagent for the following conversion?



- A. Tollen's reagent
- B. Benzoyl peroxide
- C. I_2 and NaOH solution
- D. Sn and NaOH solution

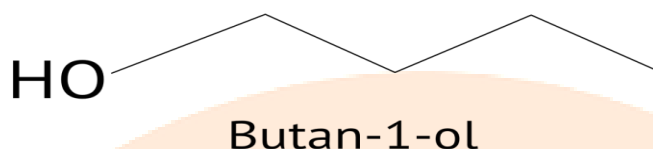
Ans: Correct option: C

I_2 and NaOH It is an iodoform reaction involving methyl ketones, the solution is the correct reagent. Because the reactants and product have double bonds, double bonds do not work throughout the reaction. There are also ketone and carboxylic groups present. The ketonic group is oxidised under the carboxylic group during the process.

11. Which of the following compounds will give butanone on oxidation with alkaline KMnO_4 solution?

- A Butan-1-ol
- B Butan-2-ol
- C Both
- D None of these

Ans: Correct option: B



When a primary alcohol, such as Butan-1-ol, reacts with a powerful oxidising agent, such as KMnO_4 , it produces carboxylic acid, however when a secondary alcohol, such as Butan-2-ol, is oxidised under the pressure of KMnO_4 , it produces ketone. As a result, option B is right.

12. In Clemmensen Reduction carbonyl compound is treated with _____.

- A. Zinc amalgam + HCl**
- B. Sodium amalgam + HCl**
- C. Zinc amalgam + nitric acid**
- D. Sodium amalgam + HNO_3**

Ans: Correct option: A

The Clemmensen Reduction method is used to reduce a carbonyl group to CH_2 . Amalgams are a type of reducing agent that has been around for a long time. In the Clemmensen reduction, hydrogen chloride acts as an acid, protonating the ketone and transferring the electrons from the zinc amalgam to the carbon atoms.

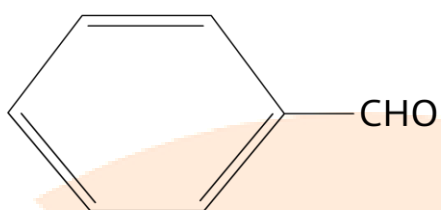
II. Multiple Choice Questions (Type-II)

Note: In the following questions two or more options may be correct

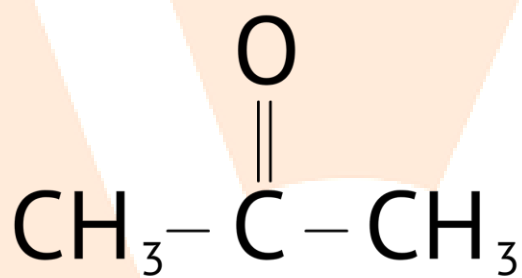
13. Which of the following compounds do not undergo aldol condensation?

A $\text{CH}_3 - \text{CHO}$

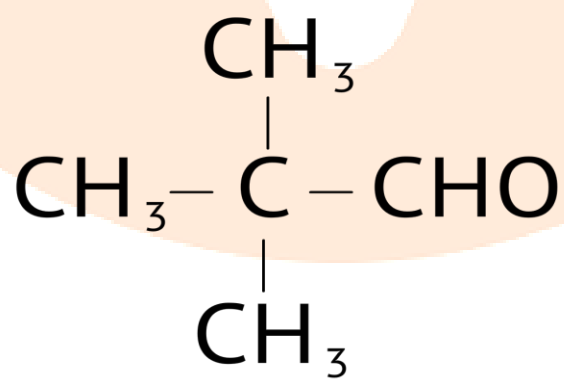
B



C



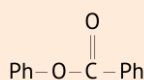
D



Ans: Correct options: B and D

Consider the following product $\text{CH}_3 - \text{CHO}$. C and O are broken into positive and negative bonds. C should have one condition: hydrogen must be available; if it isn't, aldol condensation will not occur. When two products are mixed with the availability of α -hydrogen, the positive and negative bonds between carbon and hydrogen are exchanged, and water (H_2O) is eliminated, resulting in aldol condensation.

14. Treatment of compound



with NaOH solution yields.

A Phenol

B Sodium phenoxide

C Sodium benzoate

D Benzophenone

Ans: Correct options: B and D

The carbonyl group is replaced by a methyl group. Clemmensen Reduction is used for this conversion. It's utilised to make cyclohexanone and benzophenone into cyclohexane and diphenyl methane, respectively. The reagent is a zinc-hydrochloric acid amalgam.

15. Which of the following conversions can be carried out by Clemmensen Reduction?

A Benzaldehyde into benzyl alcohol

B Cyclohexanone into cyclohexane

C Benzoyl chloride into benzaldehyde

D Benzophenone into diphenyl methane

Ans: Correct Answer: (B) and (D)

The carbonyl group of aldehydes and ketones is reduced to CH_2 group on treatment with zinc amalgam and concentrated hydrochloric acid (Clemmensen reduction) and the corresponding alkane is formed.

Only in compounds (ii) and (iv), the carbonyl group is converted into a CH_2 group.

16. Through which of the following reactions the number of carbon atoms can be increased in the chain?

- A. Grignard reaction**
- B. Cannizaro's reaction**
- C. Aldol condensation**
- D. HVZ reaction**

Ans: Correct option: A and C

Grignard Reaction is the addition of an organomagnesium halide to form a tertiary or secondary alcohol.

Canizzaro's Reaction involves the base induced disproportionation of two molecules. Aldol Condensation is an organic reaction when an enolate ion reacts with a carbonyl compound to form beta hydroxy ketone with dehydration to give a conjugated enone.

The HVZ reaction involves alpha bromination of carboxylic acids and this reacts with chlorine or bromine in presence of a small amount of red phosphorus to give alpha halo carboxylic acids.

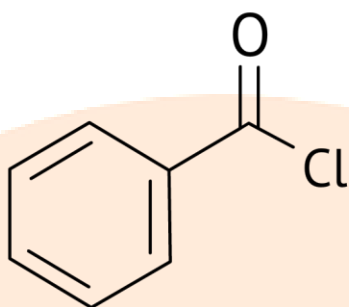
17. Benzophenone can be obtained by _____.

- A. Benzoyl chloride + Benzene + AlCl_3**
- B. Benzoyl chloride + Diphenyl cadmium**
- C. Benzoyl chloride + Phenyl magnesium chloride**
- D. Benzene + Carbon monoxide + ZnCl_2**

Ans: Correct options: A and B

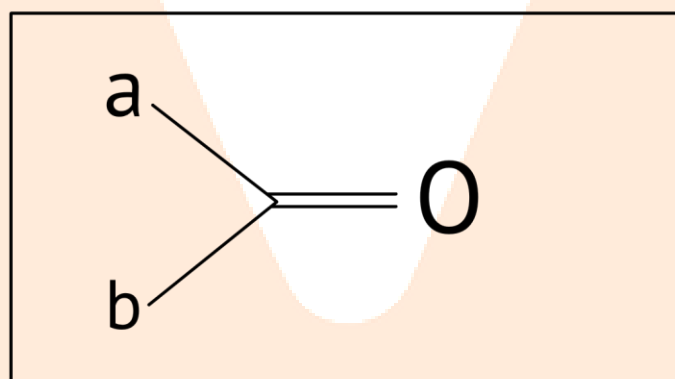
A. Benzophenone can be obtained by Friedel-craft acylation reaction

B. Benzophenone can also be obtained by the reaction between benzoyl chloride and diphenyl cadmium.

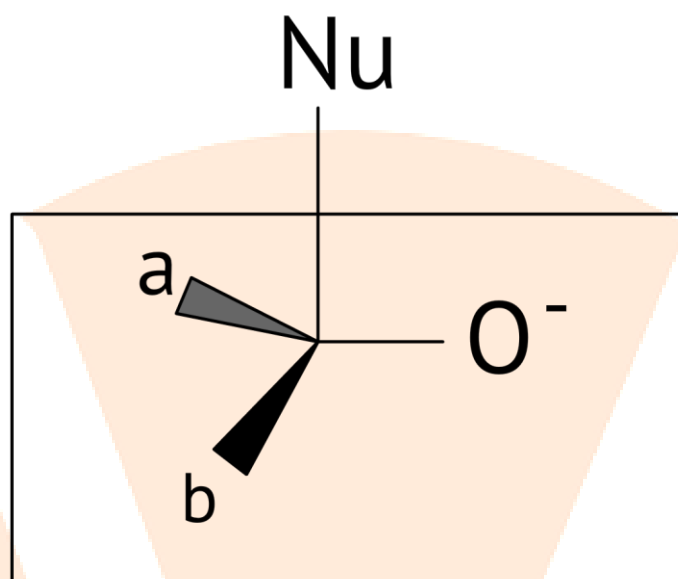


benzoyl chloride

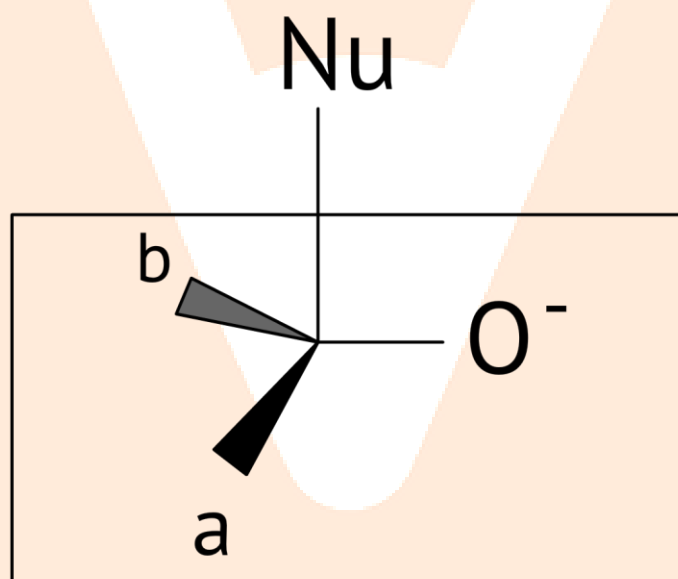
18. Which of the following is the correct representation for intermediate of nucleophilic addition reaction to the given carbonyl compound



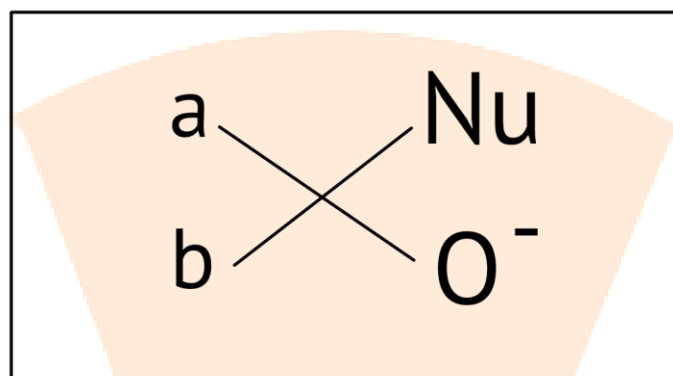
A.



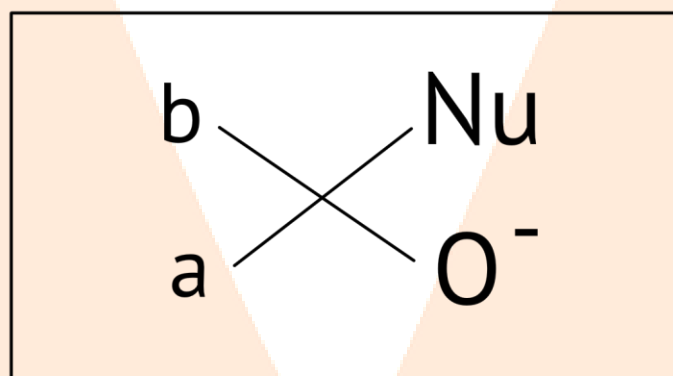
B.



C.



D.



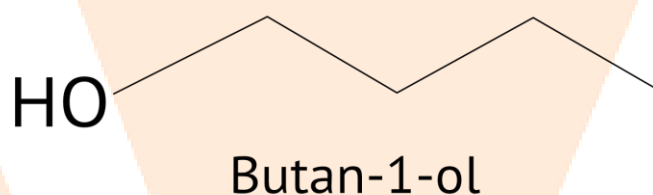
Ans: Correct options: A and B

The given compound is a planar molecule with sp^2 hybridised carbon. Carbon atoms are attacked by nucleophiles. If the nucleophile approaches from the front, the A and B molecules shift to the front and back positions, respectively, and the carbon becomes tetrahedral. Another alternative is that A and B molecules will be located above and below the plane, respectively. As a result, the choices C and D aren't represented as planar molecules.

III Short Answer Type Question

19. Why is there a large difference in the boiling points of butanal and butan-1-ol?

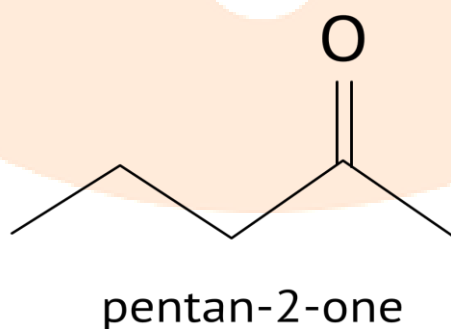
Ans:

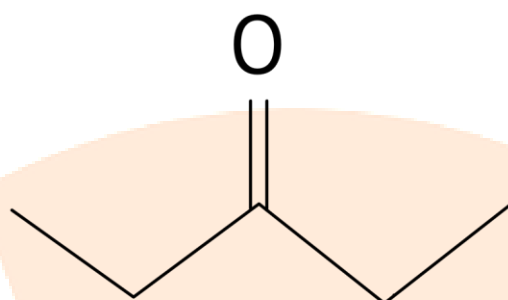


Butanol is a four-carbon alcohol, with the hydroxyl group on the first carbon. It has a functional group and a carbonyl group that interacts dipole-dipole. Because butanol possesses a polar O-H bond, it exhibits intermolecular H bonding, which is not feasible in butanal due to the lack of a polar bond. The boiling point of butanol is higher than that of butanal.

20. Write a test to differentiate between pentan-2-one and pentan-3-one

Ans:



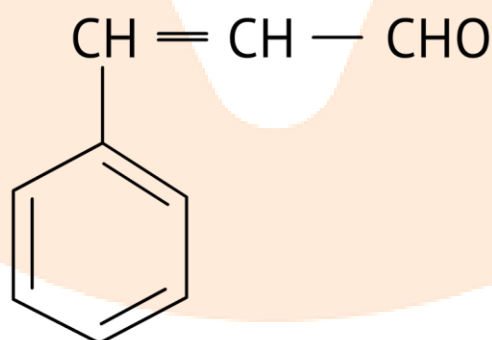


pentan-3-one

Ketone group should have methyl ketone in the iodoform test. Yellow ppt is formed when the ketone, base, and iodine react. If it's methyl ketone, the test will come back positive. Pentan-2-one and pentan-3-one are the methyl ketones.

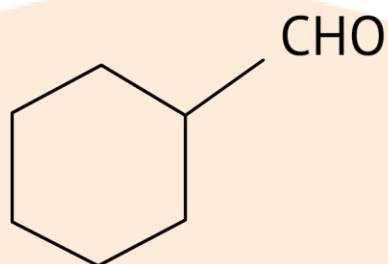
21. Give the IUPAC names of the following compounds

A.



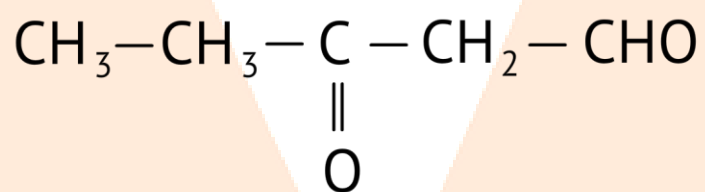
Ans: 3-Phenylprop-2-enal

B.



Ans: Cyclohexanecarbaldehyde

C.



Ans: 3-oxopentanal

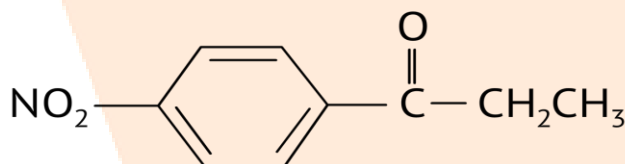
D. $\text{CH}_3 - \text{CH} = \text{CH} - \text{CHO}$

Ans: But -2-enal

22. Give the structure of the following compounds.

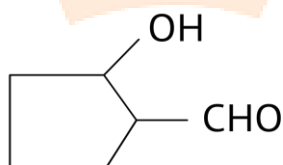
A. 4-Nitropropiophenone

Ans:



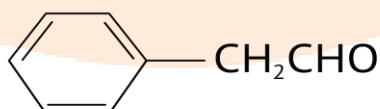
B. 2-Hydroxycyclopentanecarbaldehyde

Ans:



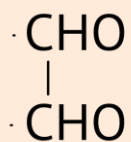
C. Phenyl acetaldehyde

Ans:



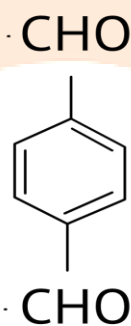
23. Write IUPAC names of the following structures.

A.



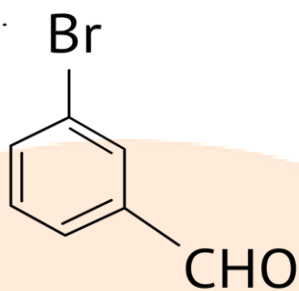
Ans: Ethane-1, 2-dial

B.



Ans: Benzene-1, 4-carbaldehyde

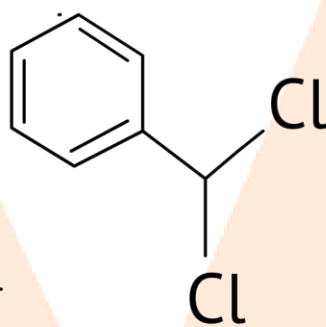
C.



Ans: 3-Bromobenzaldehyde

24. Benzaldehyde can be obtained from benzal chloride. Write reactions for obtaining benzalchloride and then benzaldehyde from it

Ans:



benzal chloride

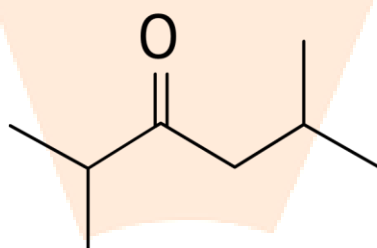
Benzal chloride can be made by chlorinating toluene in the presence of sunshine and then using the hydrolysis process to obtain benzaldehyde. This method can be used to make benzaldehyde in a commercial setting.

25. Name the electrophile produced in the reaction of benzene with benzoyl chloride in the presence of anhydrous AlCl_3 . Name the reaction also.

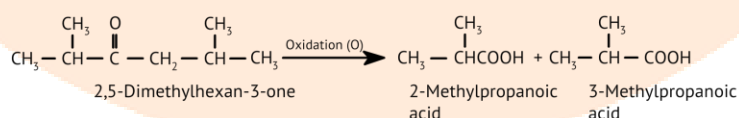
Ans: The electrophile formed when benzene reacts with benzoyl chloride in the presence of anhydrous AlCl_3 is benzoylium cation, and the resultant product is benzophenone. Friedel Crafts acylation reaction is the name for this reaction.

26. Oxidation of ketones involves carbon-carbon bond cleavage. Name the products formed on oxidation of 2, 5-dimethylhexan-3-one.

Ans:



2, 5-dimethylhexan-3-one



27. Arrange the following in decreasing order of their acidic strength and give reason for your Ans:.



Ans: The order is

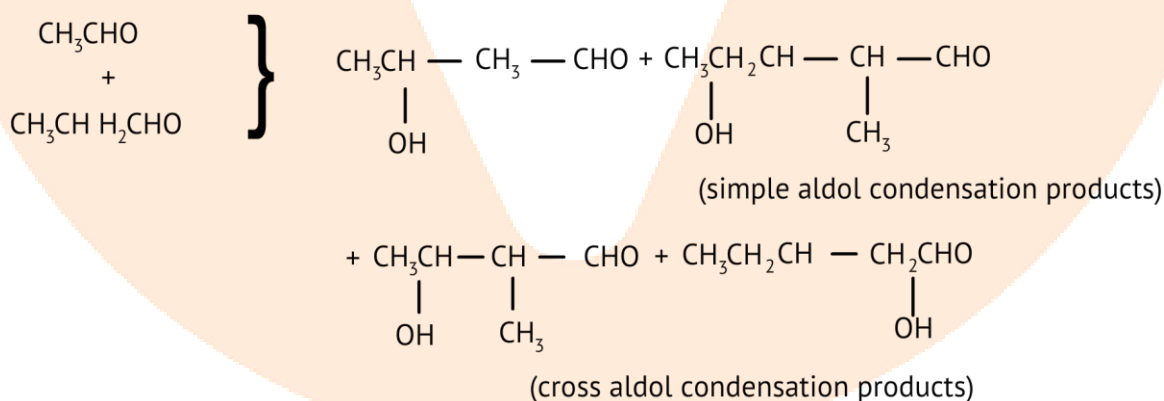


When compared to other alcohols, $\text{CH}_3\text{CH}_2\text{OH}$ is the least acidic. Because of the halogen, FCH_2COOH is quite acidic. It is either an electron donating group or an electron receiving group in acidic strength. Acidic strength increases as the number of electron withdrawing groups increases. The acidic strength of electron donating groups diminishes.

28. What product will be formed on reaction of propanal with 2-methylpropanal in the presence of NaOH? What products will be formed? Write the name of the reaction also?

Ans: It's referred to as cross Aldol Condensation.

Cross aldol condensation occurs when two separate aldehydes or ketones combine to form a single molecule.



29. Compound 'A' was prepared by oxidation of compound 'B' with alkaline KMnO_4 . Compound 'A' on reduction with lithium aluminium hydride gets

converted back to compound 'B'. When compound 'A' is heated with compound B in the presence of H_2SO_4 it produces a fruity smell of compound C to which family the compounds 'A', 'B' and 'C' belong to?

Ans: When alkaline $KMnO_4$ oxidises compound "B" with compound "A," acid is formed. $B \xrightarrow[\text{alkaline } KMnO_4]{[O]} A \rightarrow \text{acid}$

When compound "A" is reduced with $LiAlH_4$, it transforms into compound "B." The letter "B" stands for a type of alcoholic beverage.



When compounds "A" and "B" are heated with conc. H_2SO_4 , compound "C" is formed, which has a fruity odour.



30. Arrange the following in decreasing order of their acidic strength. Give for the arrangement.



Ans: $NO_2CH_2COOH > FCH_2COOH > C_6H_5COOH$

To begin, determine whether it has an acidic strength associated with it, and whether it is an electron donating or electron withdrawing group. Acidic strength increases as the number of electron withdrawing groups increases. The acidic strength reduces as the electron donating group increases. In decreasing order, the most influential acidic strength is $-NO_2 > -F > -C_6H_5$.

31. Alkenes $\begin{array}{c} \diagup \quad \diagdown \\ C = C \end{array}$ **and carbonyl compounds** $\begin{array}{c} \diagup \quad \diagdown \\ C = O \end{array}$, **both contain a pi bond but alkenes show electrophilic addition reactions whereas carbonyl compounds show nucleophilic addition reactions. Explain.**

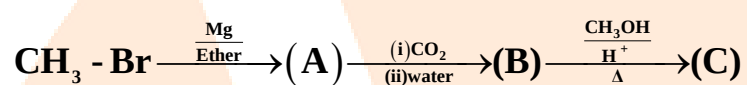
Ans: $\begin{array}{c} \diagup \quad \diagdown \\ C = O \end{array}$ carbon gains a partial positive charge, while O gains a partial negative charge. Alkenes go through an electrophilic reaction, while aldehydes

and ketones go through a nucleophilic reaction. $\begin{array}{c} \diagup \quad \diagdown \\ C = C \end{array}$ go through an electrophilic process that includes a pi bond.

32. Carboxylic acids contain carbonyl group but do not show the nucleophilic addition reaction like aldehydes or ketones. Why?

Ans: The presence of a lone pair of electrons on an atom of the OH group diminishes the electrophilic character of C through resonance; carboxylic acids contain a carbonyl group. As a result, the carbonyl atom's partial positive charge is lowered.

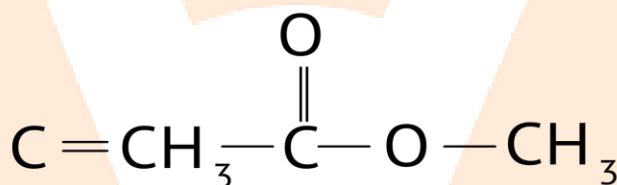
33. Identify the compounds A, B and C in the following reaction



Ans: (A) CH_3MgBr

(B) CH_3COOH

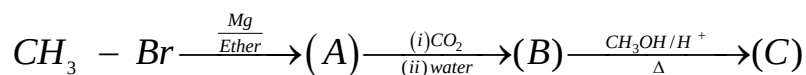
(C)



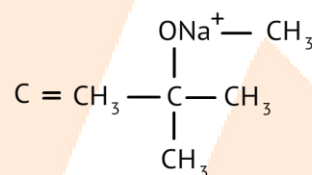
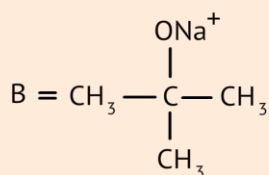
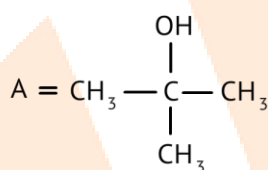
34. Why are carboxylic acids more acidic than alcohols or phenols although all of them have hydrogen atom attached to an oxygen atom (—O—H)?

Ans: Although all of them have a hydrogen atom connected to an oxygen atom (—O—H), carboxylic acids are more acidic than alcohols or phenols. This can be explained by the conjugate base's stability after H^+ ions have been removed from the acid and phenol.

35. Complete the following reaction sequence.

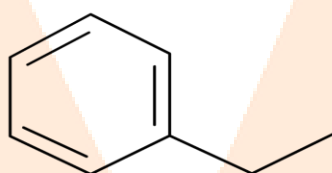


Ans:



36. Ethylbenzene is generally prepared by acetylation of benzene followed by reduction and not by direct alkylation. Think of a possible reason.

Ans:



Ethylbenzene

Due to the development of polysubstitution products, ethylbenzene is usually made through acetylation of benzene followed by reduction and direct alkylation.

Due to resonance, the benzene ring must be deactivated. The electrophilic attacking agent is rarely produced by the benzene ring.

37. Can Gatterman-Koch reaction be considered like Friedel Craft's acylation? Discuss.

Ans: Yes, both reactions look to be the same, but they are not. To convert benzene to benzaldehyde, the Gatterman-Koch reaction is performed. Benzene is treated with an acid chloride in the presence of anhydrous AlCl_3 in acylation processes. The Gatterman-Koch reaction is used to make HCOCl in situ by reacting CO with HCl gas in the presence of anhydrous AlCl_3 .

IV. Matching Type

Note: Match the items of Column I and Column II in the following questions.

38. Match the common names given in Column I with the IUPAC names given in Column II.

Match the common names given in Column I with the IUPAC names given in Column II.

Column I	Column II
Common names	(IUPAC names)
A. Cinnamaldehyde	(a) Pentanal
B. Acetophenone	(b) Prop-2-enal
C. Valeraldehyde	(c) 4-Methylpent-3-en-2-one
D. Acrolein	(d) 3-Phenylprop-2-enal
E. Mesityl oxide	(e) 1-Phenylethanone

Ans:

Column I	Column II
Common names	(IUPAC names)
A. Cinnamaldehyde	(d) 3-Phenylprop-2-enal
B. Acetophenone	(e) 1-Phenylethanone
C. Valeraldehyde	(a) Pentanal
D. Acrolein	(b) Prop-2-enal
E. Mesityl oxide	(c) 4-Methylpent-3-en-2-one

39. Match the acids given in Column I with their correct IUPAC names given in Column II

Column I	Column II
Acids	(IUPAC names)
A. Phthalic acid	(a) Hexane-1,6-dioic acid
B. Oxalic acid	(b) Benzene-1,2-dicarboxylic acid

C. Succinic acid	(c) Pentane-1,5-dioic acid
D. Adipic acid	(d) Butane-1,4-dioic acid
E: Glutaric acid	(e) Ethane-1,2-dioic acid

Ans:

Column I Acids	Column II (IUPAC names)
A. Phthalic acid	(b) Benzene-1,2-dicarboxylic acid
B. Oxalic acid	(e) Ethane-1,2-dioic acid
C. Succinic acid	(d) Butane-1,4-dioic acid
D. Adipic acid	(a) Hexane-1,6-dioic acid
E: Glutaric acid	(c) Pentane-1,5-dioic acid

40. Match the reactions given in Column I with the suitable reagents given in Column II

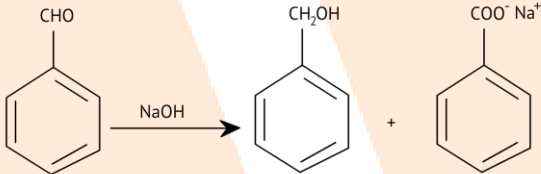
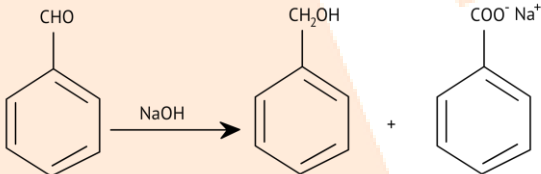
Column I (Reactions)	Column II (Reagents)
(i) Benzophenone to Diphenylmethane	(a) LiAlH_4
(ii) Benzaldehyde to 1-Phenylethanol	(b) DIBAL - H
(iii) Cyclohexanone to Cyclohexanol	(c) $\frac{\text{Zn(Hg)}}{\text{Conc.HCl}}$
(iv) Phenyl benzoate to Benzaldehyde	(d) CH_3MgBr

Ans:

Column I (Reactions)	Column II (Reagents)
(i) Benzophenone to Diphenylmethane	(c) $\frac{\text{Zn(Hg)}}{\text{Conc.HCl}}$
(ii) Benzaldehyde to 1-Phenylethanol	(d) CH_3MgBr
(iii) Cyclohexanone to Cyclohexanol	(a) LiAlH_4
(iv) Phenyl benzoate to Benzaldehyde	(b) DIBAL - H

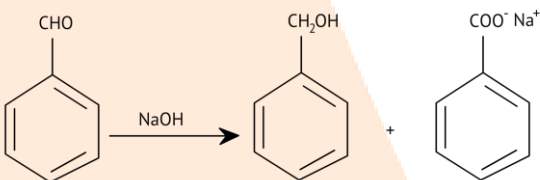
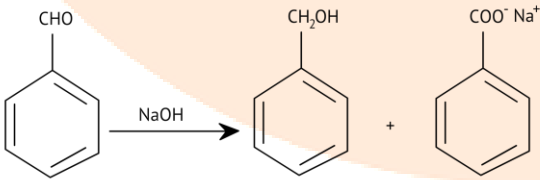
41. Match the example given in Column I with the name of the reaction in Column II

Column I	Column II
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Example	(Reaction)
A. $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl} + \text{H}_2 \xrightarrow{\text{Pd-C/BaSO}_4} \text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$	(a) Friedel Crafts acylation
B. 	(b) HVZ reaction
C. 	(c) Aldol condensation
D. $\text{R}-\text{CH}_2-\text{COOH} \xrightarrow{\text{Br}_2/\text{Red P}} \text{R}-\underset{\text{Br}}{\text{CH}}-\text{COOH}$	(d) Cannizaro's reaction
E.	(e) Rosenmund's reduction

$\text{CH}_3\text{—CN} \xrightarrow[\text{(ii) H}_2\text{O / H}^+]{\text{(i) SnCl}_2 / \text{HCl}} \text{CH}_3\text{CHO}$	
F $\text{ZCH}_3\text{CHO} \xrightarrow{\text{NaOH}} \text{CH}_3\text{—CH=CHCHO}$	(f) Stephen's reaction

Ans:

Column I Example	Column II (Reaction)
A. $\text{CH}_3\text{—}\overset{\text{O}}{\parallel}\text{C—Cl} + \text{H}_2 \xrightarrow{\text{Pd-C/BaSO}_4} \text{CH}_3\text{—}\overset{\text{O}}{\parallel}\text{C—H}$	(c) Aldol condensation
B. 	(d) Cannizaro's reaction
C. 	(a) Friedel Crafts acylation

$\text{R}-\text{CH}_2-\text{COOH} \xrightarrow{\text{Br}_2/\text{Red P}} \text{R}-\underset{\text{Br}}{\text{CH}}-\text{COOH}$	(b) HVZ reaction
D.	
$\text{CH}_3-\text{CN} \xrightarrow[\text{(ii) H}_2\text{O} / \text{H}^+]{\text{(i) SnCl}_2 / \text{HCl}} \text{CH}_3\text{CHO}$	(f) Stephen's reaction
E:	
$\text{ZCH}_3\text{CHO} \xrightarrow{\text{NaOH}} \text{CH}_3-\text{CH}=\text{CHCHO}$	(e) Rosenmund's reduction

V. Assertion and Reason Type

Note: In the following questions a statement of assertion followed by a statement of reason is given. Choose the correct Ans: out of the following choices

A. Assertion and reason both are correct, and reason is the correct of assertion.

B. Assertion and reason both are wrong statements

C. Assertion is correct statement, but reason is wrong statement.

D. Assertion is wrong statement, but reason is correct statement

E: Assertion and reason both are correct statements, but reason is not correct of assertion

42. Assertion: Formaldehyde is a planar molecule.

Reason: It contains sp^2 hybridised carbon atom

Ans: Correct answer is A.

The smallest aldehyde is formaldehyde. It features a triangular planar shape with two bond pairs and no lone pairings. C and O share one pair of electrons in their

double bond. As a result, the C atom is sp^2 hybridised. As a result, both Assertion and Reason are true.

43: Assertion: Compounds containing $-CHO$ The group is easily oxidised to corresponding carboxylic acids.

Reason: Carboxylic acids can be reduced to alcohols by treatment with $LiAlH_4$

Ans: Correct answer is E.

The two reactions produce different products for different reasons. One involves the oxidation of aldehydes, while the other involves the reduction of carboxylic acids. Aldehydes are easily oxidised by mild oxidising agents, hence compounds containing $-CHO$ are rapidly oxidised.

44. Assertion: The α -hydrogen atom in carbonyl compounds is less acidic.

Reason: The anion formed after the loss of α -hydrogen atom is resonance stabilised.

Ans: Correct answer is D.

Because of the presence of electron withdrawing carbonyl groups, the alpha hydrogen atom in carbonyl compounds is acidic. In nature, hydrogen is very acidic.

Because the cation is released in the form of H^+ , the anion created after the loss of the α -hydrogen atom is resonance stabilised.

45. Assertion: Aromatic aldehydes and formaldehyde undergo Cannizaro reaction.

Reason: Aromatic aldehydes are almost as reactive as formaldehyde

Ans: Correct answer is C.

There is no alpha hydrogen atom in the Cannizaro process. Formaldehyde and aromatic aldehydes do not contain alpha hydrogen. It proceeds through the Cannizaro reaction. Formaldehyde is the most reactive of all aldehydes.

46: Assertion: Aldehydes and ketones, both react with Tollen's reagent to form a silver mirror.

Reason: Both, aldehydes and ketones contain a carbonyl group.

Ans: Correct answer is D.

The silver mirror test can be used to determine Tollen's. $[\text{Ag}(\text{NH}_3)_2]^+ \text{OH}^-$. Only aldehydes, not ketones, react with Tollen's reagent to create silver.

A silver mirror test is not given with this affirmative test.

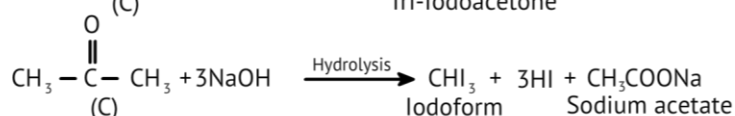
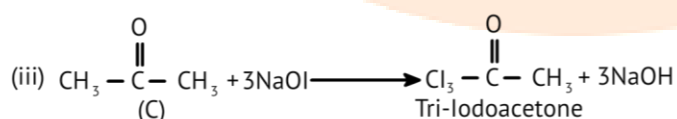
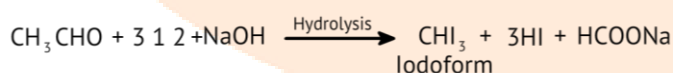
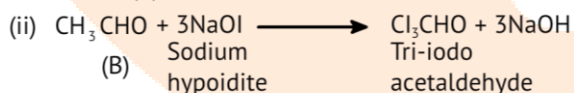
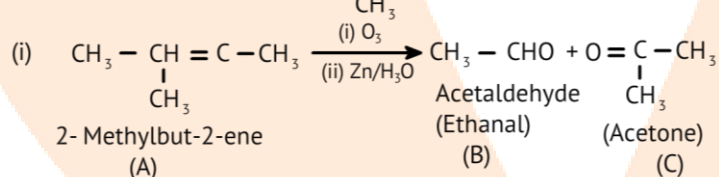
The carbonyl group is present in both aldehyde and ketone.

VI. Long Answer Type Question

47: An alkene 'A' (Mol. formula C_5H_{10} on ozonolysis gives a mixture of two compounds 'B' and 'C'. Compound 'B' gives positive Fehling's test and forms an iodoform on treatment with I_2 and NaOH . Compound 'C' does not give Fehling's test but forms an iodoform. Identify the compounds A, B and C. Write the reaction for ozonolysis and formation of iodoform from B and C.

Ans: Fehling's test is positive when compound 'B' is used. It gives an iodoform test and proves that it is an aldehyde. Because it fails Fehling's test, compound 'C' is a ketone.

Hence compound A is $\text{CH}_3 - \text{CH} = \underset{\text{CH}_3}{\text{C}} - \text{CH}_3$



48. An aromatic compound 'A' (Molecular formula C_8H_8O gives a positive 2,4-DNP test. It gives a yellow precipitate of compound B on treatment with iodine and sodium hydroxide solution. Compound A does not give Tollens or Fehlings test. On drastic oxidation with potassium permanganate it forms a carboxylic acid C (Molecular formula $C_7H_6O_2$), which is also formed along with the yellow compound in the above reaction. Identify A, B and C and write all the reactions involved.

Ans: The aromatic compound A does not give Tollens reagent test, it is not an aromatic aldehyde. As it responds to an iodoform test called methyl ketone.

49. Write down functional isomers of a carbonyl compound with molecular formula C_3H_6O . Which isomer will react faster with HCN and why? Explain the mechanism of the reaction also. Will the reaction lead to the completion with the conversion of whole reactant into product at reaction conditions? If a strong acid is added to the reaction mixture what will be the effect on concentration of the product and why?

Ans: Functional isomers of C_3H_6O containing carbonyl group are CH_3CH_2CHO and CH_3COCH_3 .

Propanol reacts faster with HCN because there is less steric hindrance and electronic factors which increases its electrophilicity. It is a reversible reaction. Addition of acid inhibits the reaction because of the formation of the $-CN$ ions. Next come the reaction mechanism

50. When liquid 'A' is treated with a freshly prepared ammoniacal silver nitrate solution, it gives bright silver mirror. The liquid forms a white crystalline solid on treatment with sodium hydrogensulphite. Liquid 'B' also forms a white crystalline solid with sodium hydrogensulphite but it does not give test with ammoniacal silver nitrate. Which of the two liquids is aldehyde? Write the chemical equations of these reactions also.

Ans: Liquid 'A' reduces ammoniacal silver nitrate and 'B' is a ketone which forms white crystalline solid on treatment with hydrogen sulphite

